

Pranav Kunnamkudath Roy

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EDUCATION

SRM Institute of Science and Technology

Kattankulathur, Tamil Nadu

B.Tech, CSE – Computer Networking

2023 – 2027

- CGPA: 8.33/10

Mount St. Mary's School

New Delhi

Class XII, CBSE

2023

- Percentage: 75.4%

Mount St. Mary's School

New Delhi

Class X, CBSE

2021

- Percentage: 91.4%

PROJECTS

RWM – Rust Window Manager | *Rust, x11rb, X11* | github.com/ReitardXd/rwm

2025

- Built a minimal tiling WM from scratch in Rust using raw X11 protocol bindings; no external WM frameworks.
- Master-stack tiling, focus-follows-mouse via EnterNotify, stack rotation, and keybinding dispatch with modifier masking.
- Graceful window kill via WM_DELETE_WINDOW protocol with XKillClient fallback; full ConfigureRequest passthrough.

Seca – Social Reading Platform | *React, Node.js, Express, MongoDB, Groq API* | github.com/ReitardXd/seca

2025

- Social reading platform where friend groups compete to finish the same book; Google OAuth login and JWT-based auth.
- Reading groups with invite codes, book search via Open Library API, and progress tracking with streaks.
- AI-generated MCQ quizzes powered by Groq (Llama 3.3); deployed on Vercel (frontend) and Render (backend).

NERVE – Network Fault Detection | *Python, Cisco DNAC API, Groq LLaMA, Streamlit, SQLite*

2025

- Automated network incident management system polling Cisco DNAC for real-time fault telemetry with remediation workflows.
- Integrated Groq LLaMA for AI-driven root cause analysis; Discord webhook alerts and Streamlit dashboard for live visualization.

N-Body Gravitational Simulator | *C++, OpenGL, GLSL, GLFW* | github.com/ReitardXd/nbody

2024

- 3D N-body simulator with modern OpenGL pipeline: bloom post-processing, particle trails, orbital camera; no physics libraries.
- Spiral galaxy procedural initial conditions; rogue black hole mode; Octree stub for Barnes-Hut $O(n \log n)$ approximation.

ProctorAI | *FastAPI, PostgreSQL, Docker, Python, Computer Vision* | github.com/ReitardXd/ProctorAI

2024

- AI-based remote proctoring platform with FastAPI backend, PostgreSQL persistence, and Docker containerization.
- Real-time gaze tracking and behavioral anomaly flagging for exam integrity monitoring during live sessions.

cracksmoker | *Python, Tkinter, hashcat, multiprocessing* | github.com/ReitardXd/cracksmoker

2024

- hashcat frontend GUI with live output streaming, mask builder, and support for MD5/SHA1/SHA256/bcrypt/SHA-512 attack modes.

TECHNICAL SKILLS

Languages: C++, C, Rust, Python, JavaScript, Bash/Shell, SQL

Systems & Networking: POSIX, Linux (Arch), TCP/IP, X11, Cisco DNAC API

Frameworks & Tools: FastAPI, Streamlit, OpenGL/GLFW, Node.js, PostgreSQL, Docker, Git, Neovim, GDB

Security: HackTheBox, CTF, hashcat, vulnerability research

Competitive Programming: Codeforces – Specialist 1590 rating

PUBLICATIONS

Smart Sleep Posture Monitoring System Using Embedded Sensors and IoT

May 2025

- International Research Journal on Advanced Engineering and Management
- ESP32-based sleep posture classifier using MPU6050 IMU and FSR sensors with ML classification and vibration feedback.

LANGUAGES

English (Native) | Hindi (Native) | Tamil (Professional Working) | Malayalam (Professional Working)